within the MTS method by a modified potential function $\bar{U}(\boldsymbol{q}) \stackrel{\text{def}}{=} U_{\text{S}}(\mathcal{A}(\boldsymbol{q}))$. The mapping $\mathcal{A}(\boldsymbol{q})$ is defined by reference to the fast dynamics, e.g. by the weighted average

$$\mathcal{A}(\boldsymbol{q}) = \frac{1}{K+1} \sum_{i=0}^{K} \phi_i \boldsymbol{\theta}_i(\boldsymbol{q}),$$

where the $\boldsymbol{\theta}_i$ are defined recursively by solving the system with Hamiltonian

$$H^{\text{fast}}(\boldsymbol{q}, \boldsymbol{p}) = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U_{\text{F}}(\boldsymbol{q})$$

starting from initial position $\boldsymbol{q}$ and zero initial momentum $\boldsymbol{p}$, for K timesteps using a small stepsize τ_f. This small step does not have to be directly related to the timestep used for integrating the fast dynamics within the MTS scheme; it is only used for the averaging procedure. The resulting sequence of $\boldsymbol{\theta}_i$ clearly are functions of $\boldsymbol{q}$ only and so $\mathcal{A}$ is well defined. The weights ϕ_i appearing in the average may be selected in various ways. The initialization of the calculation of the $\boldsymbol{\theta}_i$ can, moreover, be modified by various alternative $\boldsymbol{q}$-dependent proposals, or may be randomized.

An alternative, simpler, and probably more robust, procedure is simply to fix the value of $\mathcal{A}(\boldsymbol{q})$ by constraining $\boldsymbol{q}$ to its mechanical equilibrium value [189] in which case no fast averaging whatsoever needs to be performed. This approach relies on the use of the SHAKE method of constrained molecular dynamics (see the following section).

In each of the mollified multiple timestepping methods, since only the potential energy is modified, and by a smooth mapping of positions, the forces derived by differentiating $\bar{U}$ are still conservative, and, just as with the MTS method, the scheme remains symplectic.

The mollification is only applied for the purpose of modifying the slow impulse and the mollified methods are in other ways identical to multiple timestepping. These schemes appear to confer some distinct advantage in the large stepsize in molecular dynamics. For example where a detailed and fully flexible molecular model of a solvated biomolecule might require stepsizes of 2 fs in a Verlet simulation, mollified impulse methods may allow increases to 6 fs [189] in the 'large' timestep, i.e. the period between reevaluations of the slow forces. This benefit may be substantial in simulations of large systems.

4.3 Methods for Constrained Dynamics

Constraints are used frequently in molecular modelling. By a constraint, we mean a modification of a dynamical system to maintain the constancy of a given function of coordinates (or coordinates and momenta). In this section we consider *holonomic* constraints which may be expressed in terms of positions only, and perhaps time,